University of Geneva

Faculty of Science

Quantum Computing Lab 1

Selected Chapters

14x060

Michel Jean Joseph Donnet

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Exercise 1

Let $|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$. Write out $|\psi\rangle^{\otimes 2}$ and $|\psi\rangle^{\otimes 3}$ explicitly, both in terms of tensor products like $|0\rangle\ |1\rangle$, and using the Kronecker product.

- As we often use the abbreviated notations $|v\rangle \otimes |w\rangle = |vw\rangle$ for the tensor product, we have in term of tensor products:

$$\begin{aligned}
|\psi\rangle^{\otimes 2} &= |\psi\rangle \otimes |\psi\rangle \\
&= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}} \\
&= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} (|0\rangle \otimes |0\rangle + |0\rangle \otimes |1\rangle + |1\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle)^1 \\
&= \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2}
\end{aligned}$$

- As $|0\rangle = \binom{1}{0}$ and $|1\rangle = \binom{0}{1}$, we have by using the Kronecker[2] product:

$$\begin{aligned}
|\psi\rangle^{\otimes 2} &= |\psi\rangle \otimes |\psi\rangle \\
&= \frac{1}{\sqrt{2}}\left(\begin{pmatrix}1\\0\end{pmatrix} + \begin{pmatrix}0\\1\end{pmatrix}\right) \otimes \frac{1}{\sqrt{2}}\left(\begin{pmatrix}1\\0\end{pmatrix} + \begin{pmatrix}0\\1\end{pmatrix}\right) \\
&= \frac{1}{2}\begin{pmatrix}1\\1\end{pmatrix} \otimes \begin{pmatrix}1\\1\end{pmatrix} \\
&= \frac{1}{2}\begin{pmatrix}1\\1\\1\\1\end{pmatrix}
\end{aligned}$$

- For $|\psi\rangle^{\otimes 3}$, we have in term of tensor products:

$$\begin{aligned}
|\psi\rangle^{\otimes 3} &= |\psi\rangle \otimes |\psi\rangle^{\otimes 2} \\
&= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2} \\
&= \frac{1}{2\sqrt{2}}(|0\rangle \otimes (|00\rangle + |01\rangle + |10\rangle + |11\rangle) + |1\rangle \otimes (|00\rangle + |01\rangle + |10\rangle + |11\rangle)) \\
&= \frac{1}{2\sqrt{2}}(|000\rangle + |001\rangle + |010\rangle + |011\rangle + |100\rangle + |101\rangle + |110\rangle + |111\rangle)
\end{aligned}$$

[1] Distributivity of tensorial product [2] Kronecker product: Given vectors $\boldsymbol{a} \in \mathbb{C}^m$ and $\boldsymbol{b} \in \mathbb{C}^n$, their Kronecker product is defined as: $\boldsymbol{a} \otimes \boldsymbol{b} = (a_1\boldsymbol{b}, a_2\boldsymbol{b}, ..., a_m\boldsymbol{b})^T \in \mathbb{C}^{mn}$

- And by using the Kronecker product:

$$
\begin{aligned}
|\psi\rangle^{\otimes 3} &= |\psi\rangle \otimes |\psi\rangle^{\otimes 2} \\
&= \frac{1}{\sqrt{2}}\left(\begin{pmatrix}1\\0\end{pmatrix} + \begin{pmatrix}0\\1\end{pmatrix}\right) \otimes \frac{1}{2}\begin{pmatrix}1\\1\\1\\1\end{pmatrix} \\
&= \frac{1}{2\sqrt{2}}\left(\begin{pmatrix}1\\1\end{pmatrix} \otimes \begin{pmatrix}1\\1\\1\\1\end{pmatrix}\right) = \begin{pmatrix}1\\1\\1\\1\\1\\1\\1\\1\end{pmatrix}
\end{aligned}
$$

Exercise 2

The Hadamard operator on one qubit may be written as

$$H = \frac{1}{\sqrt{2}}[(|0\rangle + |1\rangle)\langle 0| + (|0\rangle - |1\rangle)\langle 1|].$$

Show explicitly that the Hadamard transform on n qubits, $H^{\otimes n}$, may be written as

$$H^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x,y} (-1)^{x \cdot y} \; |x\rangle\langle y|.$$

Write out an explicit matrix representation for $H^{\otimes 2}$.

- The notation $\langle\varphi|$ is used to represent the vector dual to $|\varphi\rangle$, so the vector φ conjugate and transposed.
- The notation $|\varphi\rangle\langle\psi|$ denote the matrix product between the vectors φ and ψ.

So we have:

$$
\begin{aligned}
H &= \frac{1}{\sqrt{2}}[(|0\rangle + |1\rangle)\langle 0| + (|0\rangle - |1\rangle)\langle 1|] \\
&= \frac{1}{\sqrt{2}}\left[\begin{pmatrix}1\\1\end{pmatrix}(1\ \ 0) + \begin{pmatrix}1\\-1\end{pmatrix}(0\ \ 1)\right] \\
&= \frac{1}{\sqrt{2}}\left[\begin{pmatrix}1 & 0\\1 & 0\end{pmatrix} + \begin{pmatrix}0 & 1\\0 & -1\end{pmatrix}\right] \\
&= \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\1 & -1\end{pmatrix}
\end{aligned}
$$

For $H^{\otimes 2}$, we have:

$$\begin{aligned} H^{\otimes 2} &= H \otimes H \\ &= \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \otimes \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ &= \frac{1}{2}\begin{pmatrix} 1\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} & 1\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ 1\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} & -1\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \end{pmatrix} \\ &= \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \end{aligned}$$

We want to prove that $H^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x,y} (-1)^{x \cdot y} \ |x\rangle\langle y|$ with x and y that represent n qubits, so with $|x\rangle = |x_1 x_2 ... x_n\rangle$ and $|y\rangle = |y_1 y_2 ... y_n\rangle$. So x and y are two sequences of bits.

We can define $(x \cdot y)$ as the bitwise inner product modulo 2:

$$x \cdot y = \sum_{i=1}^{n} x_i y_i \bmod 2$$

where $x_i, y_i \in \{0, 1\}$.

For the initialization, we have:

$$\begin{aligned} H^{\otimes 1} = H &= \tfrac{1}{\sqrt{2}}[(|0\rangle + |1\rangle)\langle 0| + (|0\rangle - |1\rangle)\langle 1|] \\ &= \tfrac{1}{\sqrt{2}}(|0\rangle\langle 0| + |1\rangle\langle 0| + |0\rangle\langle 1| - |1\rangle\langle 1|) \\ &= \tfrac{1}{\sqrt{2}}[(-1)^{0\cdot 0} \ |0\rangle\langle 0| + (-1)^{1\cdot 0}|1\rangle\langle 0| + (-1)^{0\cdot 1}|0\rangle\langle 1| + (-1)^{1\cdot 1} \ |1\rangle\langle 1|] \\ &= \tfrac{1}{\sqrt{2}} \textstyle\sum_{a,b\in\{0,1\}} (-1)^{a\cdot b} \ |a\rangle\langle b| \end{aligned}$$

Suppose we have

$$H^{\otimes(n-1)} = \frac{1}{\sqrt{2^{n-1}}} \sum_{x',y' \in \{0,1\}^{n-1}} (-1)^{x'\cdot y'} \ |x'\rangle\langle y'|$$

By induction,

$$
\begin{aligned}
H^{\otimes n} &= H^{\otimes(n-1)} \otimes H \\
&= \frac{1}{\sqrt{2^{n-1}}} \sum_{x',y' \in \{0,1\}^{n-1}} (-1)^{x' \cdot y'} \ |x'\rangle\langle y'| \otimes \frac{1}{\sqrt{2}} \sum_{a,b \in \{0,1\}} (-1)^{a \cdot b} \ |a\rangle\langle b| \\
&= \frac{1}{\sqrt{2^{n-1} \times 2}} \left(\sum_{x',y' \in \{0,1\}^{n-1}} (-1)^{x' \cdot y'} \ |x'\rangle\langle y'| \otimes \sum_{a,b \in \{0,1\}} (-1)^{a \cdot b} \ |a\rangle\langle b| \right) \\
&= \frac{1}{\sqrt{2^{n}}} \left(\sum_{x',y' \in \{0,1\}^{n-1}} \sum_{a,b \in \{0,1\}} (-1)^{x' \cdot y'} (-1)^{a \cdot b} \ |x'\rangle\langle y'| \otimes |a\rangle\langle b| \right)^{1} \\
&= \frac{1}{\sqrt{2^{n}}} \left(\sum_{x',y' \in \{0,1\}^{n-1}} \sum_{a,b \in \{0,1\}} (-1)^{x' \cdot y'} (-1)^{a \cdot b} \ |x'\rangle \otimes |a\rangle\langle y'| \otimes \langle b| \right) \\
&= \frac{1}{\sqrt{2^{n}}} \left(\sum_{x',y' \in \{0,1\}^{n-1}} \sum_{a,b \in \{0,1\}} (-1)^{x' \cdot y'} (-1)^{a \cdot b} \ |x'a\rangle\langle y'b| \right) \\
&= \frac{1}{\sqrt{2^{n}}} \left(\sum_{x,y \in \{0,1\}^{n}} (-1)^{x \cdot y} \ |x\rangle\langle y| \right)
\end{aligned}
$$

So we have proved by induction that

$$H^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x,y} (-1)^{x \cdot y} \ |x\rangle\langle y|.$$

Exercise 3

(Commutation relations for the Pauli matrices) Verify the commutation relations

$$[X, Y] = 2iZ; \quad [Y, Z] = 2iX; \quad [Z, X] = 2iY.$$

There is an elegant way of writing this using ε_{jkl}, the antisymmetric tensor on three indices, for which $\varepsilon_{jkl} = 0$ except for $\varepsilon_{123} = \varepsilon_{231} = \varepsilon_{312} = 1$, and $\varepsilon_{321} = \varepsilon_{213} = \varepsilon_{132} = -1$:

$$\left[\sigma_j, \sigma_k\right] = 2i \sum_{l=1}^{3} \varepsilon_{jkl} \sigma_l.$$

The commutation relation is defined as follow:

$$[A, B] = AB - BA$$

The Pauli matrix are defined as follow:

[1] We can write this thanks to the bilinearity of the tensor product: $(X_1 + X_2) \otimes (Y_1 + Y_2) = \sum_{i,j} X_i \otimes Y_j$

$$\sigma_0 \equiv I \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\sigma_1 \equiv \sigma_x \equiv X \equiv \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\sigma_2 \equiv \sigma_y \equiv Y \equiv \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\sigma_3 \equiv \sigma_z \equiv Z \equiv \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

So we have:

- $$\begin{aligned} [X, Y] &= XY - YX \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} - \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} \\ &= \begin{pmatrix} 2i & 0 \\ 0 & -2i \end{pmatrix} \\ &= 2i \times \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ &= 2iZ \end{aligned}$$
- $$\begin{aligned} [Y, Z] &= YZ - ZY \\ &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 2i \\ 2i & 0 \end{pmatrix} \\ &= 2i \times \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= 2iX \end{aligned}$$

- $$\begin{aligned}
[Z, X] &= ZX - XZ \\
&= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix} \\
&= 2 \times \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \\
&= 2 \times \begin{pmatrix} 0 & (-i)^2 \\ i^2 & 0 \end{pmatrix} \\
&= 2iY
\end{aligned}$$

Exercise 4

Suppose $[A, B] = 0$, $\{A, B\} = 0$, and A is invertible. Show that B must be 0.

The anticommutation relation is defined as follow:

$$\{A, B\} = AB + BA$$

So if we have $[A, B] = 0$, $\{A, B\} = 0$, and A invertible, it means that

$$\begin{aligned}
&[A, B] = \{A, B\} \\
\Leftrightarrow\ & AB - BA = AB + BA \\
\Leftrightarrow\ & 2BA = 0 \\
\Leftrightarrow\ & BAA^{-1} = 0A^{-1} \\
\Leftrightarrow\ & B = 0
\end{aligned}$$

So B must be 0.

Exercise 5

Suppose A and B are commuting Hermitian operators. Prove that $\exp(A)\exp(B) = \exp(A + B)$.

An operator is Hermitian if its transpose conjugate is equal to the operator itself.

As A and B are commuting, we have $[A, B] = 0 \Leftrightarrow AB - BA = 0 \Leftrightarrow AB = BA$.

The Taylor expansion of the exponential function for an operator X is:

$$\exp(X) = \sum_{k=0}^{\infty} \frac{X^k}{k!}$$

So we have:

$$\begin{aligned}
\exp(A)\exp(B) &= \sum_{k=0}^{\infty} \frac{A^k}{k!} \sum_{j=0}^{\infty} \frac{B^j}{j!} \\
&= \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} \frac{A^k B^j}{k!j!} \\
&= \sum_{n=0}^{\infty} \sum_{k=0}^{n} \frac{A^k B^{n-k}}{k!(n-k)!} && \text{let} \quad n = j + k \\
&= \sum_{n=0}^{\infty} \sum_{k=0}^{n} \frac{n!}{n!k!(n-k)!} \left(A^k B^{n-k}\right) \\
&= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} \left(A^k B^{n-k}\right) \\
&= \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{k=0}^{n} \binom{n}{k} \left(A^k B^{n-k}\right) && \text{by definition of } \binom{n}{k} = \frac{n!}{k!(n-k)!} \\
&= \sum_{n=0}^{\infty} \frac{1}{n!} (A+B)^n && \text{according to binomial theorem}^1 \\
&= \exp(A+B) && \text{according to taylor expansion}
\end{aligned}$$

Exercise 6

Suppose we have a qubit in the state $|0\rangle$, and we measure the observable X. What is the average value of X? What is the standard deviation of X?

The state of a qubit can be described as the combination of two states: $|0\rangle$ and $|1\rangle$. So the state of a qubit $|\varphi\rangle$ can be written as

$$|\varphi\rangle = \alpha\ |0\rangle + \beta\ |1\rangle$$

with $\alpha, \beta \in \mathbb{C}$.

As the qubit state is a combination of two states, the qubit state cannot be observed. However, the qubit state can be measured to fix the state of the qubit to state $|0\rangle$ or state $|1\rangle$.

According to the Born Rule, for a qubit $|\varphi\rangle = \alpha\ |0\rangle + \beta\ |1\rangle$, the probability that the qubit is in the state $|0\rangle$ is given by $|\alpha|^2$ and the probability that the qubit is in the state $|1\rangle$ is given by $|\beta|^2$. As $|\alpha|^2$ and $|\beta|^2$ are probabilities, we have that $|\alpha|^2 + |\beta|^2 = 1$.

If we consider that we have a qubit in the state $|0\rangle$, it means that $|\alpha|^2 = 1$ and $|\beta|^2 = 0$ since $|\alpha|^2 + |\beta|^2 = 1 \Leftrightarrow 1 + |\beta|^2 = 1 \Leftrightarrow |\beta|^2 = 0 \Leftrightarrow \beta = 0$. As α is a complex number, we cannot determine the value of α.

So the average value of the observable X is given by:

[1] Binomial theorem: $(A+B)^n = \sum_{k=0}^{n} \binom{n}{k} A^k B^{n-k}$

$$\mathbb{E}[X] = \alpha \; |0\rangle + \beta \; |1\rangle = \alpha \; |0\rangle$$

The standard deviation of X is given by:

$$\begin{aligned}
\mathrm{Var}[X] &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \\
&= \mathbb{E}\big[(\alpha \; |0\rangle + \beta \; |1\rangle)^2\big] - (\alpha \; |0\rangle)^2 \\
&= \mathbb{E}\left[\alpha^2 \begin{pmatrix}1\\0\end{pmatrix} \begin{pmatrix}1\\0\end{pmatrix}^T\right] - \alpha^2 \begin{pmatrix}1\\0\end{pmatrix} \begin{pmatrix}1\\0\end{pmatrix}^T \\
&= \alpha^2 \begin{pmatrix}1\\0\end{pmatrix} \begin{pmatrix}1\\0\end{pmatrix}^T - \alpha^2 \begin{pmatrix}1\\0\end{pmatrix} \begin{pmatrix}1\\0\end{pmatrix}^T \\
&= 0
\end{aligned}$$

Exercise 7

Calculate the probability of obtaining the result $+1$ for a measurement of $\vec{v} \cdot \vec{\sigma}$, given that the state prior to measurement is $|0\rangle$. What is the state of the system after the measurement if $+1$ is obtained?

Exercise 8

Show that the average value of the observable $X_1 Z_2$ for a two qubit system measured in the state $(|00\rangle + |11\rangle)/\sqrt{2}$ is zero.

Exercise 9

Verify that the Bell basis forms an orthonormal basis for the two qubit state space.

Exercise 10

Suppose E is any positive operator acting on Alice's qubit. Show that $\langle\psi|E \otimes I|\psi\rangle$ *takes the same value* when $|\psi\rangle$ is any of the four Bell states. Suppose some malevolent third party ('Eve') intercepts Alice's qubit on the way to Bob in the superdense coding protocol. Can Eve infer anything about which of the four possible bit strings $00, 01, 10, 11$ Alice is trying to send? If so, how, or if not, why not?

To archive all results (solutions, report together with your code, etc.) and submit them using Quantum computing course moodle.